SYSDESIGN REFERENCE

CAP Theorem Tradeoff Matrix

A decision-oriented comparison of alternatives and failure modes.

During a network partition, a distributed operation must choose between a consistent response and an available response.

VERSION 1.0 REVIEW

REVIEWED THROUGH 2026-07-18

STATUS HUMAN REVIEW REQUIRED

SYSDESIGN / CAP-THEOREM

SYSDESIGN

CAP Theorem Tradeoff Matrix

CORE IDEA

During a network partition, a distributed operation must choose between a consistent response and an available response.

Reference architecture decision chain

01	02	03	04
Define The Operation	Consistency Guarantee	Failure Model	And User-Visible Fallback.

Decisions to defend

+ Define the operation
+ consistency guarantee
+ failure model
+ and user-visible fallback.

Failure modes to test

+ Treating CAP as a permanent two-of-three menu or ignoring latency and normal-operation tradeoffs.

Interview flow

- [ ] Clarify users, critical paths, and non-goals
- [ ] Estimate only scale that changes a decision
- [ ] Define API, data model, and source of truth
- [ ] Draw synchronous and asynchronous paths
- [ ] Pressure-test timeout, retry, overload, and recovery
- [ ] Close with tradeoffs and one alternative

Deep-dive drill

CAP Theorem Tradeoff Matrix

Probe	What a strong answer makes explicit
How does the design behave during a regional dependency failure?	Requirement, decision, cost, failure behavior, and observable metric.
Which metric reveals the first bottleneck?	Requirement, decision, cost, failure behavior, and observable metric.
What consistency does the user actually observe?	Requirement, decision, cost, failure behavior, and observable metric.
How would the design change at ten times the traffic?	Requirement, decision, cost, failure behavior, and observable metric.

Data questions

+ Partition key
+ Versioning
+ Uniqueness
+ Retention
+ Reconciliation

Operations questions

+ SLO
+ Backlog
+ Capacity
+ Alert
+ Failover test

DESIGN DISCIPLINE

For every box: which requirement justifies it, what cost does it introduce, and what happens when it fails?

PUBLISHING STATUS

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